

# Citation under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

Harsh DUBEY<sup>1</sup>

*New York University, New York, USA*

**Abstract.** Most research on fabricated legal citations scores the errors in finished briefs; this paper measures how they are produced. Seven models from seven vendors, open-weight 30B to frontier, drafted argument sections for 48 U.S. Supreme Court matters under five prompting conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 drafts was checked deterministically, with no language-model judge: existence against an 18-million-row reporter index, quotations against a verbatim checker validated on seeded faults and audited by hand on real drafts, and, for quotations on U.S. Reports authority, pinpoint pages against official reporter pagination from public archives. The failure mode moves with capability: under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, verbatim quotation accuracy fell for four models in the pre-registered fit, three of them under both templates (six of seven in an exploratory fit pooling three generations), the count of accurate quotations per draft fell for 4 models, two of them at the top of the range, and Sonnet 5’s rate did not fall, most plausibly because it quoted less under pressure; its residual errors are paraphrase presented as quotation, real language from the wrong case, and wrong pages. In a second experiment the checker fed its findings back into revision. Sonnet 5 reached 1.000 verbatim accuracy over the quotations it kept: of its 36 flagged quotations, 3 were corrected and 31 deleted, and accurate quotations per draft rose for one model of seven. A flag carrying the closest passage of the cited opinion raised corrections fourfold, and removal still exceeded repair in every model. A verifier with 50 percent precision removed 20 of Sonnet 5’s 68 correct quotations, as many as revision with no feedback (21) and against 3 under a precise one. A second task on 48 federal appellate matters shows lower baseline existence for six of seven models; instruments, drafts, and records are released.

**Keywords.** legal citation, hallucination, large language models, verification, evaluation

## 1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs with citations a language model invented, from *Mata v. Avianca* [1] to a docket a public tracker now

---

<sup>1</sup>Corresponding Author: Harsh Dubey, New York University; hd2225@nyu.edu.

counts in the thousands [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model or agent find [3,4]? Detection is the right measure for reviewing a filed brief, not for asking which prompting conditions produce the errors, this paper’s question.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and for which models? Deployed prompts carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a publication-year window lowered citation existence even in capable models; we add a sanctions warning and know of no comparable study for law. Second, when a deterministic checker feeds its findings back into revision, does the model correct them, leave them, or change the draft so that the tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the best published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall), which its authors attribute to page-level information living behind commercial databases [3]. We built it from public sources: existence against a local index derived from CourtListener bulk data, verbatim quotation checking validated on seeded faults and audited on real drafts, and page-level pinpoint verification for quotations on U.S. Reports authority against the official pagination in the Caselaw Access Project’s volume archives [6].

Our primary setting is the closed-book use of general-purpose models, the setting of most sanctioned filings documented to date [2]; a grounded arm with ten retrieved, verified authorities in the prompt measures what retrieval changes. Drafts under the four non-baseline conditions are the pressured drafts (1,344 in the Supreme Court task, three refusals). The paper contributes a constraint-sensitive evaluation, in which reliability at a neutral prompt does not predict reliability under drafting constraints and the failure mode moves with capability; deterministic verification of existence, verbatim quotation, and pinpoint page with no language-model judge, released with the drafts; and the finding that verification is not repair, since flags fed back to the model raise measured accuracy by deletion and an imprecise verifier damages correct content.

## 2. Related Work

**Legal citation reliability.** Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their error categories and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book for a named case’s citation almost never produce it; our task asks whether the cases a model chooses to cite exist. Dahl et al. [7] and Magesh et al. [8] measure hallucinated authority in chatbots and research tools.

**Constraints and fabrication.** Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified by fuzzy bibliographic matching that left 36 to 61 percent of citations unresolved. We transplant their factorial to law with deterministic verification (0.2 percent unresolved) and a formal statistical model.

**Revision with execution feedback.** The loop protocol follows execution-feedback work in code generation [9], with its no-feedback control separating feedback content from extra revision; we add controls in which a quarter to all of the feedback is false.

**AI and law.** Blair-Stanek and Van Durme [10] found that frontier models flip the winning party across identical zero-temperature runs, which Section 8 bounds with repeated generations. Evaluation in this community pairs automated metrics with the checks lawyers perform [11,12]; citation integrity is one with deterministic ground truth. Garneau et al. [13] found retrieving the paragraph a proposition rests on far harder than retrieving the case; our verifier checks a generated pincite against the page text. Zhang et al. [14] show models miss overruled precedent, ConsRAG [15] constrains retrieval, and Trozze et al. [16] and Harasta et al. [17] study how readers perceive model-written documents. Evaluations of generated legal argument [18,19] do not check the authority it cites; citation-network work [20,21,22,23] studies the network our checker queries.

### 3. Tasks, Conditions, and Instruments

**Terms.** In 347 U.S. 483, 495, the second page is the *pincite*, for a quotation the page where the quoted language appears. *Citation existence* is the share of adjudicable citations that resolve to a real case; *strict quotation accuracy* the share of scored quotations, those the draft attributes to a case whose text the checker holds, that reproduce the opinion verbatim; a quotation the draft does not attribute is outside the rate.

**Matters.** The matters are forty-eight Supreme Court cases from 1990 to 2020 drawn from the Supreme Court Database [24]. Each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, the Supreme Court’s decision excluded and outcome leakage screened. The model argues an assigned side in an argument section of under 900 words; nothing predicts outcomes [25]. A second task uses forty-eight published F.3d decisions from the Caselaw Access Project archive (1995 to 2019, four from each of the First through Eleventh and Federal Circuits, no cross-appeals); the packet is the opinion’s statement of the case and facts, cut before the analysis and scrubbed of outcome sentences, and the model argues for the appellant under all five conditions.

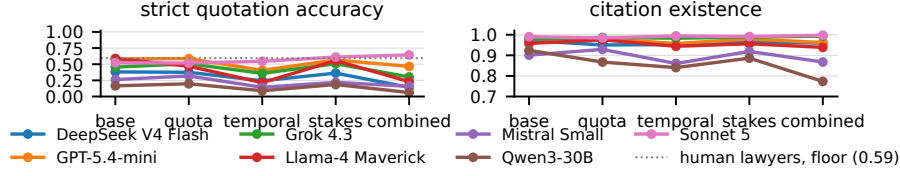
**Conditions.** The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”), a clause no user writes, a checkable stand-in for the authority limits practitioners impose; unlike a jurisdiction limit, which pushes toward the circuit authority where the second task finds fabrication, it moves citations toward well-memorized opinions, the gentler direction; stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which concatenates the three clauses. Each condition has one fixed template in the primary run, at temperature 0. The stakes clause names Rule 11, the rule invoked in the sanctioned-filing cases [1], which governs district-court civil actions [26] and neither task’s forum, so the clause is a stakes signal only. A second template rephrasing every condition was run on all 48 matters and seven models. A *grounded* arm reran baseline and combined with ten retrieved authorities in the prompt: the U.S. Reports opinions most similar to the question and facts (TF-IDF over 25,250 cached opinions decided before argument and, for the combined condition, before 1970, neither party named), each with citation, year, and its best-matching 250-word passage, stated to be real, quotation restricted to the excerpts.

**Models.** The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor;

tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are called the top of the range. The seven were run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens (identifiers in the repository). An eighth model, GLM-4.7-flash, exhausted a 30,000-token budget in reasoning on 33 of 240 drafts and is excluded.

**Adjudication.** Citations are extracted with *eyecite* [27] and given one of three labels: exists, not found, or unverifiable (vendor citations and coverage gaps), so that index coverage is never counted as fabrication; 0.2 percent of the 14,040 citation records in the full run are unverifiable. Citations resolve against an 18-million-row table; quotations are located by balanced quotation marks and attributed, in order, to the citation whose parenthetical holds them, the case named in their own sentence, the citation that follows them in that sentence (not one introduced by *see* or *cf.*), the citation an *Id.* points to, or the last citation in the paragraph; a quotation introduced by a statute, rule, the record, or a lower court, or with no citation in its paragraph, is left unpaired, since the checker has nothing to test it against: 37 percent of the 10,846 quotations in the full run, 24 percent introduced by non-case material, 10 percent standing in a paragraph that cites nothing, and 3 percent too far from any citation, a share that runs from 31 to 42 percent across models. *Exists* means the volume, reporter, and page resolve to an indexed opinion (13,241 of the 14,040 records; 774 are not found); for the 11,769 whose draft names the parties the case name is compared separately, and 205 (1.7 percent; 1.1 at baseline, 2.8 under the temporal clause; 0.3 for Sonnet 5, 3.5 for Mistral Small) carry a name the index does not match, abbreviated names included; counting every mismatch as not found keeps the 3 surviving existence contrasts, adds Mistral Small's temporal fall (OR 0.59, corrected  $p=0.041$ ), and removes Sonnet 5's uncorrected combined rise (OR 1.07). A quotation is *near miss* when every fragment reaches token coverage of at least 0.85 without matching exactly, and *inaccurate* otherwise. A lenient rate counting near misses as correct is reported beside the strict rate. Matching is case and punctuation insensitive, splits on ellipses and on bracketed alterations (so that a lawful substitution is read as an omission and the words around it must match), and drops alteration parentheticals, though an internal citation omitted without an ellipsis still fails; the lenient rate is a diagnostic, since 0.85 coverage admits one wrong word in seven. The quotation checker scores 240 of 240 on twenty synthetic drafts planting twelve item kinds, six of them attribution traps, and the existence and pinpoint checks 100 of 100 on 20 seeded opinions. Pinpoint pages are verified against the official reporter pagination in the Caselaw Access Project's HTML volumes, which mark every page break; a pincite range is read at its first page, a quotation is located by its longest fragment, so one that straddles a page break is placed where that fragment sits, and only quotations whose fragment is somewhere in the cited case enter the page-level rate.

**Human reference.** The same instrument scored 482 pre-ChatGPT appellate brief excerpts from Liu et al. [3] (no injected error; OCR-converted, so a floor); human lawyers reach 0.594 strict and 0.776 lenient quotation accuracy (416 scored quotations) and 0.957 citation existence, a reference for the severity of the verbatim standard only, since human briefs differ in citation mix. The literal matcher the strict standard replaced, which read a bracketed alteration as its contents, puts the human rate at 0.493 and moves model cells by at most 9 points, the same 8 contrasts surviving; 59 of the 169 human strict failures are near misses without any alteration, the signature of a corrupted character.



**Figure 1.** Strict quotation accuracy (left) and citation existence (right) by condition, one line per model; the dotted line is the human reference of Section 3, a severity floor on OCR-converted briefs, not a comparison; the right axis starts at 0.7.

**Table 1.** Baseline versus the combined condition; quotation columns give the strict rate over attributed quotations (lenient) [scored], existence columns the rate [not found / adjudicable], the last column the share of checked citations decided after 1969 under the temporal condition; an asterisk marks a contrast surviving Holm correction within model.

| Model        | Quotes, base        | Quotes, comb.        | Exist, base    | Exist, comb.    | Viol., temporal |
|--------------|---------------------|----------------------|----------------|-----------------|-----------------|
| Qwen3-30B    | 0.165 (0.609) [230] | 0.061 (0.320) [325]* | 0.924 [28/370] | 0.773 [90/397]* | 20%             |
| Mistral      | 0.262 (0.620) [187] | 0.157 (0.444) [248]  | 0.901 [37/373] | 0.868 [69/521]  | 35%             |
| DeepSeek     | 0.381 (0.730) [252] | 0.142 (0.491) [281]* | 0.972 [10/354] | 0.959 [20/488]  | 11%             |
| Grok 4.3     | 0.455 (0.737) [156] | 0.302 (0.647) [119]  | 0.981 [5/257]  | 0.994 [3/479]   | 0%              |
| Sonnet 5     | 0.523 (0.804) [260] | 0.644 (0.825) [194]  | 0.991 [3/318]  | 0.998 [1/450]   | 0%              |
| GPT-5.4-mini | 0.583 (0.819) [127] | 0.466 (0.693) [88]   | 0.981 [8/417]  | 0.962 [27/713]  | 13%             |
| Llama-4      | 0.588 (0.784) [51]  | 0.230 (0.557) [61]*  | 0.958 [9/213]  | 0.939 [31/506]  | 10%             |

**Inference.** The primary analysis is a citation-level logistic generalized estimating equation per model,  $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$  for citation  $i$  in matter  $m$  under condition  $k$ , with exchangeable working correlation within matters [28], so that the 48 matters are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected  $p$  values within model, each model being the family the paper draws conclusions about, and call an effect significant only when it survives correction; counts of models are descriptive, and the effective sample is 48 matters, not thousands of citations. A matter-level cluster bootstrap agreed in sign with every corrected result. Every other fit is exploratory and follows the same model and the same rule, Holm within model over that fit’s own tests: the lenient outcome, distinct authorities, the second template, and pooled templates, tasks, and generations (Section 8), each pooling with its fixed effect; the released matrix gives every contrast under every fit. For the revision experiment, pre-registered descriptively, we add matter-paired comparisons of final strict accuracy between arms (Wilcoxon, bootstrap interval). Hypotheses and decision rules were frozen before the full run; every later decision, including two withdrawn findings, is in a dated ledger.

#### 4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

**Citation existence.** In the single run, only the 30B model’s existence moves significantly under pressure: from 0.924 at baseline to 0.773 under the combined condition,

with the quota, temporal, and combined contrasts all significant after correction (combined OR 0.30, 95 percent CI 0.18 to 0.51, corrected  $p < 0.001$ ). Mistral Small falls without reaching significance, and the five other models stay between 0.93 and 1.00 in every condition. Each surviving existence contrast rests on at least 28 citations not found per arm; a record is one occurrence of a full citation (short forms are not counted), and refitting on distinct authorities per draft leaves the same 8 contrasts.

**Quotation accuracy.** The temporal restriction damages quoting most; after correction, strict quotation accuracy falls under temporal and combined pressure for Llama-4 (0.588 to 0.230), under the combined condition for Qwen3-30B (0.165 to 0.061) and DeepSeek (0.381 to 0.142), and under the temporal clause for Mistral Small; GPT-5.4-mini and Grok 4.3 fall in the same direction without surviving correction in the single run, and Sonnet 5 does not fall (below). Llama-4 shows the largest drop in this fit, from 0.588 at baseline to 0.214 under the temporal restriction (OR 0.18, CI 0.08 to 0.43, corrected  $p < 0.001$ ), on cells of 51 and 42 scored quotations, and it does not replicate under the second template (below); refitting every contrast on the lenient outcome, 8 survive correction, the combined fall among them for Qwen3-30B, DeepSeek, and Mistral Small. Old, scanned opinions are not the cause: baseline strict accuracy is 0.439 on pre-1970 opinions (189 quotations) and 0.383 on later ones (1,074). With three generations pooled, the combined-condition fall survives correction for 6 of the seven models (OR 0.20 to 0.53), Sonnet 5 excepted, and Mistral Small's existence fall also survives (OR 0.62, corrected  $p=0.004$ ).

**Accurate quotations per draft.** The strict rate is conditional on quoting: scored quotations per cell run from 42 to 325, the unpaired share from 15 to 50 percent. We therefore also test, post hoc, the count of accurate quotations per draft, Wilcoxon paired by matter ( $n=48$ ), Holm within model over four conditions. The count falls from baseline to combined for all seven models, and the fall survives correction for 7 contrasts in 4 models, all under the temporal or combined clause: DeepSeek (2.00 to 0.83 per draft), Llama-4 (0.62 to 0.29), and two models whose rate contrasts did not survive, GPT-5.4-mini (1.54 to 0.85, corrected  $p=0.033$ ) and Grok 4.3 (1.48 to 0.75,  $p=0.015$ ). Sonnet 5's count is flat (2.83 to 2.60,  $p=0.532$ ) while its inaccurate quotations per draft fall from 2.58 to 1.44.

**Robustness to phrasing.** Under the second template, 12 contrasts survive correction, among them 5 of the 8 that survived under the first: Qwen3-30B's existence falls under the temporal and combined clauses and its quotation fall under the combined, DeepSeek's combined and Mistral Small's temporal quotation falls, the five effects that hold under both templates; Qwen3-30B's quota existence fall and Llama-4's two quotation falls did not replicate, and seven contrasts, all falls, survive under the second template only (GPT-5.4-mini's temporal existence; quotation for DeepSeek and Qwen3-30B under temporal, GPT-5.4-mini, Grok 4.3, and Mistral Small under combined, Mistral Small under stakes). Pooling both templates, all 8 survive and the combined-condition fall in quotation accuracy survives for 6 of seven models (OR 0.27 to 0.46), Sonnet 5 again the exception (pooled OR 1.28, corrected  $p=0.963$ ).

**Second task: federal appellate briefs.** Baseline citation existence on the 48 appellate matters runs from 0.652 for Qwen3-30B to 0.993 for Sonnet 5, lower than on the Supreme Court matters for 6 of the seven models, Sonnet 5 alone unchanged. The failures concentrate in the federal reporters: 22 percent of citations to F.2d, F.3d, and F. Supp. do not resolve, against 4 percent for the U.S. Reports; the index resolves all but 3 of the 759

federal-reporter citations in the deciding opinions themselves, so the shortfall is not coverage. One pressure contrast survives correction within the task, Llama-4’s existence fall under the temporal clause (0.950 to 0.857, OR 0.32, corrected  $p=0.018$ ). Pooling both tasks with a task effect, seven of the 8 Supreme Court survivors remain (Qwen3-30B’s quota existence fall drops out), joined by four further falls and by Sonnet 5’s rate rise under the stakes clause (OR 1.53, corrected  $p=0.020$ ).

**Grounded drafting.** With ten verified authorities in the prompt, citations not found fell from 341 in 5,856 closed-book to 74 in 5,605 (Qwen3-30B combined existence 0.773 to 0.995, OR 53.8, CI 14.7 to 197.2,  $p<0.001$ ); quotation accuracy rose in every cell yet, with the source passages in the prompt, no cell exceeded 0.734, and 61 percent of Sonnet 5’s 71 remaining inaccurate quotations are paraphrases of the cited case in quotation marks.

**The strongest model does not degrade; the leading explanation is that it quotes less.** Sonnet 5’s strict rate is numerically higher under the temporal, stakes, and combined conditions and highest under the combined one (0.523 to 0.644; uncorrected  $p = 0.0313$ , corrected  $p = 0.250$ ), the one pre-registered contrast in that direction and one that reaches no fit. Sonnet 5 wrote 9.6 quotations per draft at baseline and 6.1 under the combined condition; its accurate quotations per draft held (2.83 to 2.60) while its inaccurate ones fell (2.58 to 1.44), so the quotations it stopped writing were disproportionately the ones that would have failed. Selection is the leading reading; which quotations were dropped, by length or case age, is not tested, and the temporal restriction also shifts the cited cases toward older, better-memorized opinions (median decision year 1943 against 1986). With three generations pooled Sonnet 5’s existence rise survives the logistic fit (corrected  $p=0.006$ ) but rests on twelve citations not found (7 of 975 against 5 of 1,324), on which an exact test gives  $p=0.38$ ; the quotation rise reaches corrected  $p=0.404$  there. In the appellate task the stakes clause alone raises the rate (0.480 to 0.620, corrected  $p=0.084$ ) without moving the cited cases (median year 1991 in both); the direction is consistent across tasks.

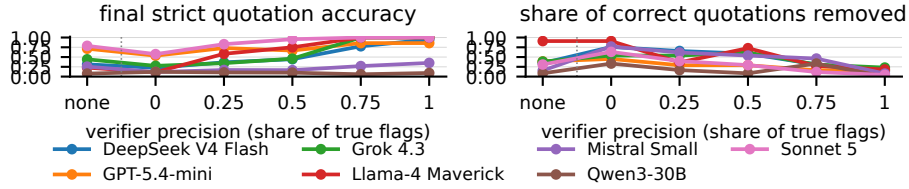
## 5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop: each round the checker returned one line per failure, the model revised, and the loop ended on a clean draft or after three revisions. A *false feedback* control gave the same number of identically formatted lines aimed at correct items, *mixed* controls replaced each line with a false one at random for a precision of about 0.25, 0.5, or 0.75, and a *no-feedback* control gave the same rounds with a generic instruction to revise. The loop used the first 24 matters in pre-registered order; loop rates are means over drafts with at least one scored quotation (68 of the 168 true-arm episodes ended with none, 19 of Grok 4.3’s 24, so its final rate rests on 5 drafts), so round-0 figures differ from Table 1. The loop was rerun after the attribution rule was corrected; the original run, whose true-arm flags had realized precision 0.71 under the corrected rule (0.38 for Sonnet 5), is released beside it. The rerun’s flags are the corrected rule’s own verdicts, whose precision on quotation flags is 78 percent by the hand reading of Section 6 (71 to 84), so the precision axis of Figure 2 is nominal and its arm labelled 1 sits nearer 0.8. Every scored round-0 quotation was traced to the final draft (Table 2).

**Satisfying the checker by deletion.** Under true feedback the final strict rate ranges from 0.092 for Qwen3-30B to 1.000 for Sonnet 5 (Table 2). Sonnet 5 had 36 flagged

**Table 2.** Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft; corr.: an accurate quotation on the same citation now stands in its place; left: still present and flagged; removed: deleted, dequoted, or made inaccurate; added: new in the final draft, accurate/flagged; strict accuracy at round 0 and final as the draft mean, with the rate pooled over quotations in parentheses, round 0 / final; Clean: episodes ending on a passing draft, with quotations / with none left.

| Model        | Flagged at round 0 |       |          |              | Accurate at round 0 |         |       | Strict, r0 → final (pooled)      | Clean |
|--------------|--------------------|-------|----------|--------------|---------------------|---------|-------|----------------------------------|-------|
|              | n                  | corr. | dequoted | deleted left | n                   | removed | added |                                  |       |
| Qwen3-30B    | 138                | 0     | 10       | 13           | 115                 | 12      | 0     | 0/32 0.089 → 0.092 (0.080/0.075) | 0/0   |
| Mistral      | 96                 | 4     | 4        | 43           | 45                  | 13      | 1     | 1/27 0.233 → 0.350 (0.119/0.191) | 3/7   |
| DeepSeek     | 136                | 2     | 59       | 71           | 4                   | 29      | 4     | 1/2 0.292 → 0.950 (0.176/0.824)  | 11/12 |
| Grok 4.3     | 37                 | 0     | 23       | 14           | 0                   | 13      | 3     | 0/0 0.234 → 1.000 (0.260/1.000)  | 5/19  |
| Sonnet 5     | 36                 | 3     | 2        | 31           | 0                   | 68      | 3     | 0/0 0.737 → 1.000 (0.654/1.000)  | 23/1  |
| GPT-5.4-mini | 22                 | 1     | 1        | 19           | 1                   | 31      | 5     | 0/1 0.538 → 0.857 (0.585/0.931)  | 11/9  |
| Llama-4      | 33                 | 0     | 5        | 28           | 0                   | 11      | 2     | 0/0 0.273 → 1.000 (0.250/1.000)  | 6/18  |



**Figure 2.** Nominal verifier precision (share of true feedback lines; leftmost point, no feedback) against final strict quotation accuracy (left) and the share of accurate round-0 quotations that revision deleted, dequoted, or made inaccurate (right).

quotations at round 0: 1 was corrected in place, 2 replaced by a new accurate quotation from the same case, 2 dequoted, and 31 deleted, while only 3 of its 68 accurate quotations went, against 21 under revision with no feedback. Qwen3-30B left most flags standing and Mistral Small nearly half (45 of 96); the other five removed most of what was flagged. Counted over all 24 episodes, the intent-to-treat measure, accurate quotations per draft rose for 1 of seven models (Mistral Small, 0.54 to 0.71) and were flat or lower for the rest (Sonnet 5 2.83 to 2.83, Grok 4.3 0.54 to 0.42), so the loop raised the rate over surviving quotations and not the number of correct ones in a draft. Citation existence rose by the same route: of 113 citations unresolved at round 0, 82 were gone from the final draft, 31 remained, and 37 new ones resolved. Rerun from the grounded combined drafts, the loop repaired 4 of 23 flags for Sonnet 5 and 1 of 51 for DeepSeek. A *passage* arm then put the evidence in the flag itself: each quotation line carried the closest passage of the cited opinion (Section 3’s best-matching passage). Repair rose in six of seven models (Grok 4.3 still repaired none): to 12 of 36 for Sonnet 5, 26 of 96 for Mistral Small against 4; removal still exceeded it in every model (23 for Sonnet 5).

**Controls: feedback content and verifier precision.** Without feedback (Figure 2), Sonnet 5’s strict rate still rose (0.737 to 0.787), and true feedback added 0.21 over revision alone (paired over the 23 matters scored in both arms; interval 0.12 to 0.32,  $p=0.002$ ), as for DeepSeek (0.38,  $n=12$ ); GPT-5.4-mini’s, Qwen3-30B’s, and Mistral Small’s differences (0.04, 0.02, 0.05;  $n=11, 23, 16$ ) are within noise; Llama-4 and

Grok 4.3 left too few paired matters. Sonnet 5's correct quotations removed rise as precision falls, 3, 8, 20, 27, and 43 of 68 at nominal precisions of 1, 0.75, 0.5, 0.25, and 0, against 21 with no feedback: at one half the verifier removed what revision alone removes, and the damage beyond that appears at 0.25 and 0. Under all-false feedback five of seven models ended below their starting rate while flagged counts fell in every model: what changed was the deletion of correct quotations.

## 6. Residual Errors at the Top of the Range

**Paraphrase inside quotation marks.** Decomposing every inaccurate quotation against the cited opinion's text, 68 percent of Sonnet 5's (169 of 250) are paraphrases of the correct case presented as verbatim quotation, 56 match nothing substantial in the cited opinion, and 25 cite authority outside the U.S. Reports.

**Real language, wrong case.** Searching every inaccurate quotation against the 62,049 U.S. Reports opinions in the Caselaw Access Project cache locates 326 quotations across the seven models (72 for Sonnet 5, 43 for Qwen3-30B) whose language appears verbatim in a different opinion than the one cited (139 with the true source cited in the same draft, so a ceiling). By provenance, Sonnet 5's 250 partition into 72 verbatim in another opinion, 130 found nowhere in the corpus, 43 matching only stock phrases, and 5 present in the attributed case after all, the search's bound on the checker's error. A hand reading of 160 inaccurate verdicts drawn at random across models found 35 on the checker's side (22 percent, 95 percent interval 16 to 29: non-case material in quotation marks, or a quotation bound to the wrong case), 111 the model's, 4 failing only on a bracketed alteration, and 10 open. The share was 8 of 28 at baseline, lowest under the combined condition (4 of 42) and highest under the quota (11 of 30), and by model runs from 14 percent (Llama-4, 1 of 7) to 27 (Qwen3-30B, 12 of 45); carried into the fit by reclassifying every inaccurate verdict as unpaired at its condition's share (200 draws), all 8 pre-registered survivors survive in every draw, and in at least 99 percent of draws at the upper end of each condition's interval, since the error is smallest where the falls are. 40 accurate verdicts read the same way all held; the rule's first version put about half of a forty-item sample on the checker's side. In the grounded arm, where the cited passages were in the prompt, a reading of 40 inaccurate verdicts found 14 on the checker's side (35 percent, 22 to 50: 6 statutory or policy text, 4 a lower court's words, 4 bound to the wrong case, one through a token-boundary fault in the name rule the ledger records), a higher share because the model's own failures are rarer there.

**Right case, wrong page.** Of the 9,129 pinpoints in the Supreme Court drafts, 73 percent were checked against the cited case's page span (the rest cite other reporters or cases past volume 572, the archive's last) and 1,006 carry a quotation on the same citation that the cited case contains somewhere, the page-level rate's scope; for those the quotation lies on the cited page 83 percent of the time for Sonnet 5, 50 to 67 percent for the other six, and 88 percent for the human briefs; 12 percent lie on an adjacent page and the rest elsewhere in the cited case.

## 7. Discussion

**Where the checker belongs.** Two recommendations follow from the loop results; no arm had a person in it, so they are inferences, not findings. The first is the checker at the gate, returning failures to a person, over a loop in which the model resolves bare flags, and, if a loop is built, the passage in the flag: removing an unverifiable quotation is a defensible response to a flag, but false flags get the same response and a loop-pattern tool reports a clean draft.

**Verifier precision is a deployment requirement.** Our checker’s error is 22 percent by the hand reading; the best published detection agent reaches 84.4 percent recall at an F1 of 55.0, a precision near 41 percent [3]. At precision one half Sonnet 5 lost 20 of 68 correct quotations, no more than revision with no feedback removed (21), and the damage beyond that appears at 0.25 and 0 (Figure 2). A pipeline that feeds a detector’s flags to a model should report the detector’s precision beside its recall. The second is to report the strict standard as defined here, which admits bracketed alterations and which the OCR-converted human reference passes at 0.594, with 0.85 coverage as a diagnostic.

**What evaluations should include.** The two experiments make one point. A model can raise an observable correctness rate by changing what it exposes to the check, quoting less under pressure (Section 4) and deleting under feedback (Section 5), without the underlying quoting improving; a rate should be read beside the count of correct items it is computed over.

## 8. Limitations

Both tasks are United States federal case law, the stakes clause names a rule that governs neither forum, and the checks stop at existence, wording, and page, since three model panels for relevance, calibrated on expert labels [3], fell short of a 0.75 bar, so none was used [29,30]. The controls, grounded arm and loop, second task, trace, count outcome, and corrected attribution rule postdate the pre-registered analysis and are reported apart from it. Three generations of the baseline and combined cells bound run-to-run variation [10]: cell standard deviations were at most 0.050 (existence) and 0.057 (strict), and the combined-versus-baseline quotation contrast kept its sign in every run for all 7 models; the between-template spread (15 points) exceeds many single-run effects, so we headline only contrasts that survive across templates. The seven models are one per vendor, so capability, family, and recipe are confounded.

## 9. Conclusion

Under a quota, a date restriction, and a sanctions warning the failure mode moves with capability: the 30B model loses existence, the middle of the range loses verbatim quotation and correct quotations per draft, and the strongest model’s rate holds, most plausibly because it quotes less. Fed back to the model, the checker’s flags are removed far more often than corrected, even with the passage in the flag, and false flags remove correct quotations; we recommend verification at the gate and verifier precision in the evaluation of any system that acts on it. Instruments, drafts, records, and ledger are at <https://github.com/hardy30894/citation-under-pressure>.

## References

- [1] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); 2023. Opinion and order on sanctions, No. 1:22-cv-01461 (PKC), June 22, 2023.
- [2] Charlotin D. AI Hallucination Cases Database; 2025. Accessed 2 September 2026; 2,008 cases listed. <https://www.damiencharlotin.com/hallucinations/>.
- [3] Liu P, Stambach D, Henderson P. Who Checks the Citations? Benchmarking Legal Hallucination Detection; 2026. Accepted at the Conference on Language Modeling (COLM) 2026. Available from: <https://arxiv.org/abs/2606.21155>. doi:10.48550/arXiv.2606.21155.
- [4] Chen S, Yin H, Zhou S. LegalCiteBench: Evaluating Citation Reliability in Legal Language Models; 2026. Available from: <https://arxiv.org/abs/2605.10186>. doi:10.48550/arXiv.2605.10186.
- [5] Zhao C, Tang Y, Qian Y. Do Deployment Constraints Make LLMs Hallucinate Citations? An Empirical Study across Four Models and Five Prompting Regimes; 2026. Available from: <https://arxiv.org/abs/2603.07287>. doi:10.48550/arXiv.2603.07287.
- [6] Harvard Law School Library Innovation Lab. Caselaw Access Project; 2024. Digitized collection of official, book-published U.S. case law; accessed September 2026. <https://case.law/>.
- [7] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*. 2024;16(1):64-93. doi:10.1093/jla/laae003.
- [8] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*. 2025;22(2):216-42. doi:10.1111/jels.12413.
- [9] Gehring J, Zheng K, Copet J, Mella V, Cohen T, Synnaeve G. RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning. In: *Proceedings of the 42nd International Conference on Machine Learning*. vol. 267 of *Proceedings of Machine Learning Research*. PMLR; 2025. p. 19034-55. Available from: <https://proceedings.mlr.press/v267/gehring25a.html>.
- [10] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Chicago, IL, USA: ACM; 2025. p. 425-9. doi:10.1145/3769126.3769245.
- [11] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 86-97. doi:10.3233/FAIA251579.
- [12] Sanchi M, Novotná T. Towards Hybrid Evaluation Methodologies for Large Language Models in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 389-92. doi:10.3233/FAIA241279.
- [13] Garneau N, Olsen HP, Corduant A, Tarissan F. Combining Network and Text to Provide Legal Pincites. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 282-7. doi:10.3233/FAIA241256.
- [14] Zhang L, Savelka J, Ashley KD. Do LLMs Truly “Understand” when a Precedent Is Overruled? In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 239-50. doi:10.3233/FAIA251592.
- [15] Nguyen HT, Satoh K. ConsRAG: Minimize LLM Hallucinations in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 327-32. doi:10.3233/FAIA241263.
- [16] Trozze A, Davies T, Kleinberg B. Large language models in cryptocurrency securities cases: can a GPT model meaningfully assist lawyers? *Artificial Intelligence and Law*. 2025;33(3):691-737. doi:10.1007/s10506-024-09399-6.
- [17] Harasta J, Novotná T, Savelka J. It cannot be right if it was written by AI: on lawyers’ preferences of documents perceived as authored by an LLM vs a human. *Artificial Intelligence and Law*. 2026;34(1):153-90. doi:10.1007/s10506-024-09422-w.
- [18] Gray MA, Savelka J, Oliver WM, Ashley KD. Generating Legal Arguments with Automatically Identified Factor Magnitudes. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 199-208. doi:10.1145/3769126.3769212.
- [19] Pires R, Malaquias Junior R, Nogueira R. Automatic Legal Writing Evaluation of LLMs. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 420-4. doi:10.1145/3769126.3769227.

- [20] Khatri M, Sheik R, Wadhwa P, Satija G, Kumar Y, Shah RR, et al. CiteCaseLAW: Citation Worthiness Detection in Caselaw for Legal Assistive Writing. In: Legal Knowledge and Information Systems: JURIX 2023, vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 257-62. doi:10.3233/FAIA230971.
- [21] Sadeghian A, Sundaram L, Wang DZ, Hamilton WF, Branting K, Pfeifer C. Automatic semantic edge labeling over legal citation graphs. Artificial Intelligence and Law. 2018;26(2):127-44. doi:10.1007/s10506-018-9217-1.
- [22] Dadgostari F, Guim M, Beling PA, Livermore MA, Rockmore DN. Modeling law search as prediction. Artificial Intelligence and Law. 2021;29(1):3-34. doi:10.1007/s10506-020-09261-5.
- [23] Mumford J, Florimonte F, Atkinson K, Dzehtsiarou K. Context-Aware Citation Networks: A Human-AI Dataset, Analysis, and Tool. In: Legal Knowledge and Information Systems: JURIX 2025, vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 145-55. doi:10.3233/FAIA251584.
- [24] Spaeth HJ, Epstein L, Nelson MJ, Martin AD, Segal JA, Ruger TJ, et al. Supreme Court Database, Version 2025 Release 01; 2025. Modern database, terms 1946 to 2024. <https://scdb.la.psu.edu/>.
- [25] Ortega JE, Joshi DD, Borkowski MP. Large-Language Memorization During the Classification of United States Supreme Court Cases; 2025. Available from: <https://arxiv.org/abs/2512.13654>. doi:10.48550/arXiv.2512.13654.
- [26] Federal Rules of Civil Procedure, Rule 1 (Scope and Purpose); 2024. "These rules govern the procedure in all civil actions and proceedings in the United States district courts". [https://www.law.cornell.edu/rules/frcp/rule\\_1](https://www.law.cornell.edu/rules/frcp/rule_1).
- [27] Cushman J, Dahl M, Lissner M. eyecite: A tool for parsing legal citations. Journal of Open Source Software. 2021;6(66):3617. Available from: <https://doi.org/10.21105/joss.03617>. doi:10.21105/joss.03617.
- [28] Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. Biometrika. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
- [29] Norman JD, Rivera MU, Hughes DA. Reliability without Validity: A Systematic, Large-Scale Evaluation of LLM-as-a-Judge Models Across Agreement, Consistency, and Bias; 2026. Available from: <https://arxiv.org/abs/2606.19544>. doi:10.48550/arXiv.2606.19544.
- [30] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena; 2023. NeurIPS 2023 Datasets and Benchmarks Track. Available from: <https://arxiv.org/abs/2306.05685>. doi:10.48550/arXiv.2306.05685.